RB-312: DNS Resolution Failures in Kubernetes

Field	Value
Service	CoreDNS and NodeLocal DNSCache (namespace kube-system, cluster prod-eu-1)
Severity	SEV-2 (SEV-1 if more than 25% of lookups fail)
Owner	Kubernetes Platform team
Last reviewed	2026-10-21

1. Symptoms

- Alert CoreDNSErrorsHigh fires when SERVFAIL responses exceed 1% of all CoreDNS responses for 5 minutes.
- Alert CoreDNSLatencyHigh fires when CoreDNS p99 response time is above 250 ms for 10 minutes.
- Alert CoreDNSNxdomainSpike fires when the NXDOMAIN rate is more than 5x its 7-day baseline.
- Applications log getaddrinfo EAI_AGAIN or dial udp 169.254.20.10:53: i/o timeout.

2. How DNS resolution works in the cluster

1. A pod sends its query to NodeLocal DNSCache on its own node at 169.254.20.10.
2. On a cache miss, node-local-dns forwards cluster names to the kube-dns service at 10.96.0.10, served by 6 CoreDNS replicas.
3. CoreDNS forwards external names to the VPC resolvers 10.0.0.2 and 10.0.0.3, run by Edge Networking.

Pods use ndots:5 by default, so a name with fewer than 5 dots, such as api.stripe.com, is first tried with each of the 4 search domains. Every external lookup therefore produces 4 NXDOMAIN answers before the real one.

3. Diagnosis

3.1 Test from a debug pod

```
kubectl run dnsdebug --rm -it --image=registry.internal/netshoot:0.12 -- bash
dig +search orders-svc.orders.svc.cluster.local
dig @10.96.0.10 api.stripe.com
```

If queries to 10.96.0.10 work but default lookups fail, the problem is node-local-dns on that node (section 3.5).

3.2 Check CoreDNS health

```
kubectl -n kube-system get pods -l k8s-app=kube-dns -o wide
kubectl -n kube-system logs -l k8s-app=kube-dns --tail=200 | grep -E "SERVFAIL|i/o timeout"
```

i/o timeout towards 10.0.0.2 means the VPC resolvers are failing: page Edge Networking. CPU above 80% of the 1-core limit on every CoreDNS pod means more replicas are needed.

3.3 Investigate NXDOMAIN spikes

In Grafana, break coredns_dns_responses_total{rcode="NXDOMAIN"} down by query name. Names ending in .svc.cluster.local with an external domain inside, such as api.stripe.com.payments.svc.cluster.local, are the ndots:5 search-path effect from one noisy workload.

3.4 Check conntrack exhaustion

DNS over UDP uses one conntrack entry per query. On the affected node:

```
sudo sysctl net.netfilter.nf_conntrack_count net.netfilter.nf_conntrack_max
sudo conntrack -S | grep -o "insert_failed=[0-9]*"
```

The limit is 262144 entries. A rising insert_failed count explains timeouts on a single node.

3.5 Check NodeLocal DNSCache

```
kubectl -n kube-system get ds node-local-dns
kubectl -n kube-system get pods -l k8s-app=node-local-dns --field-selector spec.nodeName=<NODE>
```

The DaemonSet must show DESIRED equal to READY. A pod in CrashLoopBackOff breaks DNS for every pod on that node.

4. Remediation

4.1 Scale CoreDNS

Run kubectl -n kube-system scale deployment coredns --replicas=10. Scale back to 6 once the incident is over; the permanent replica count is managed by Kubernetes Platform.

4.2 Lower ndots for the noisy workload

Ask the owning team to add dnsConfig.options with ndots: "2" to the pod spec, or to call external names with a trailing dot (api.stripe.com.). Either removes the 4 extra NXDOMAIN lookups.

4.3 Raise the conntrack limit

As an immediate fix on the affected node only:

```
sudo sysctl -w net.netfilter.nf_conntrack_max=524288
```

A manual sysctl is lost on reboot; Kubernetes Platform sets the permanent value in the node pool configuration.

4.4 Restart node-local-dns on one node

Run kubectl -n kube-system delete pod <NODE_LOCAL_DNS_POD>; the DaemonSet recreates it within about 20 seconds. Restart one node at a time.

Warning: do not delete the node-local-dns DaemonSet and do not edit the coredns ConfigMap by hand. Pods use 169.254.20.10 as their only nameserver, and the Corefile is managed by the Argo CD app coredns-prod.

5. Verification

- dig from the debug pod answers in under 10 ms for cluster names and under 50 ms for external names.
- CoreDNSErrorsHigh and CoreDNSLatencyHigh resolve, and insert_failed stops increasing.

6. Escalation

Level	Who	Contact
L1	On-call SRE	#ops-oncall
L2	Kubernetes Platform on-call (CoreDNS, node-local-dns, conntrack)	PagerDuty: k8s-platform
L3	Edge Networking on-call (VPC resolvers, external zones)	PagerDuty: edge-net